

1. Administrative & Study Identification

Full Study Title: Efficacy and Safety of Tisagenlecleucel in Adult Patients With Refractory or Relapsed Follicular Lymphoma
Short Title/Acronym: ELARA **Protocol Number:** [Internal Study ID / Novartis Protocol] **NCT Number:** NCT03568461 **Posting ID:** 32808306644895204 **Sponsor/Company Name:** Novartis Pharmaceuticals Corporation **Investigational Product:** Tisagenlecleucel (CTL019, Kymriah®) - CD19-directed chimeric antigen receptor (CAR) T cell therapy [ATC Code: L01XL04] **Phase of Development:** Phase II **Trial Status:** COMPLETED **Medical Monitor:** [Name and Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To determine the efficacy of tisagenlecleucel by evaluating the Complete Response Rate (CRR) based on the Lugano 2014 classification criteria as determined by an Independent Review Committee (IRC). * **Secondary Objectives:** To evaluate Overall Response Rate (ORR), Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), pharmacokinetics, cellular kinetics, immunogenicity, and safety profile.

2.2 Background & Rationale Follicular lymphoma (FL) is the second most common non-Hodgkin lymphoma in the Western hemisphere. There are limited, curative treatment options for patients who are refractory to or relapse after standard systemic therapies (such as anti-CD20 antibodies and alkylating agents). Tisagenlecleucel, an autologous anti-CD19 CAR-T cell therapy, previously demonstrated a highly durable complete remission rate in a pilot study of relapsed/refractory B-cell lymphomas. The ELARA trial was designed to verify these benefits in a larger, global, adult patient population with relapsed or refractory FL.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm * **Blinding:** Open-label * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Target \$\$\$:** 98 adult patients * **Number of Sites:** Multicenter, global trial * **Estimated Study Start:** [DD-MMM-YYYY] * **Estimated Primary Completion:** [DD-MMM-YYYY]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Refractory or relapsed Follicular Lymphoma (Grade 1, 2, 3A). 3. Radiographically measurable disease at screening. 4. Refractory to or relapsed after ≥ 2 lines of systemic therapy (must include an anti-CD20 antibody and an alkylating agent), or relapsed after autologous hematopoietic stem cell transplant (auto-SCT).

4.2 Exclusion Criteria 1. Evidence of histologic transformation. 2. Follicular Lymphoma Grade 3B. 3. Prior anti-CD19 therapy. 4. Prior gene therapy. 5. Prior adoptive T cell therapy. 6. Prior allogeneic hematopoietic stem cell transplant. 7. Active CNS involvement by malignancy. 8. Active or serious uncontrolled infections.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Single intravenous (IV) infusion with a target dose of 0.6×10^8 CAR-positive viable T cells. * **Route of Administration:** Intravenous Infusion. * **Duration of Treatment:** Single infusion event followed by extensive short-term and long-term efficacy and safety follow-up.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Optional bridging chemotherapy prior to infusion, and mandatory lymphodepleting (LD) chemotherapy. * **Prohibited:** Concurrent investigational therapies and systemic immunosuppressive medications that could interfere with CAR-T cell expansion and persistence.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening/Pretreatment	Days prior to infusion
2	Treatment	Day 0
3	Early Follow-up	Day 28 & Month 3
4	Long-term Follow-up	Months 6 - 24+
5	Survival Follow-up	Q3 Months (Post-24M)

Determine survival status and disease progression every 3 months

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Complete Response Rate (CRR) – defined as the percentage of participants with a best overall response (BOR) of complete response (CR) recorded from infusion until progressive disease or start of new anticancer therapy. *

Measurement Method: Determined by an Independent Review Committee (IRC) using CT and PET scans based on the Lugano 2014 classification criteria. Anatomical measurements include index/non-index/new lesions and spleen length. PET response is based on a 5-point scale (5PS) or Deauville score.

7.2 Secondary & Exploratory Endpoints * Overall Response Rate (ORR: % of CR + PR) Per IRC Assessment * Duration of Response (DOR) Per IRC Assessment * Progression-Free Survival (PFS) * Overall Survival (OS) * Cellular kinetics, immunogenicity (antibody titers specific to tisagenlecleucel), and minimal residual disease (MRD) status

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring specifically for Grade $\geq$ 3 cytokine release syndrome (CRS), immune effector cell-associated neurotoxicity syndrome (ICANS), severe infections, fatigue, and cytopenias.
 - **Serious Adverse Events (SAEs):** Reported per standard regulatory timelines (e.g., 24 hours for life-threatening events).
 - **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee evaluates safety signals continuously throughout the trial.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Efficacy Set (evaluable infused patients) and Safety Set (all patients who received the tisagenlecleucel infusion).
 - **Sample Size Justification:** Powered to demonstrate a statistically significant and clinically meaningful improvement in CRR compared to historical control data.
 - **Handling of Missing Data:** Standard survival analysis methodologies (Kaplan-Meier estimates for DOR, PFS, and OS); patients lost to follow-up are censored at the date of last contact.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine onsite and remote monitoring of trial sites to ensure protocol adherence, CAR-T handling logistics, and accurate AE reporting.
 - **Audit Trail:** Stringent eCRF tracking for all cellular product chain-of-identity and chain-of-custody logs from leukapheresis through manufacturing to patient infusion.
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11. Study Identifiers & Governance

Primary ID: ELARA **Registry Number:** NCT03568461 **Posting ID:** 32808306644895204 **Sponsor/Lead Organization:** Novartis Pharmaceuticals Corporation **Study Title:** Efficacy and Safety of Tisagenlecleucel in Adult Patients With Refractory or Relapsed Follicular Lymphoma **Official Regulatory Title:** A Phase II, Single Arm, Multicenter Open Label Trial to Determine the Efficacy and Safety of Tisagenlecleucel (CTL019) in Adult Patients With Refractory or Relapsed Follicular Lymphoma

12. Clinical Framework (Oncology/Lymphoma Specific)

Disease Area: Follicular Lymphoma **Primary Condition:** Relapsed or Refractory Follicular Lymphoma (r/r FL) **Diagnosis Threshold:** Radiographically measurable, Grades 1, 2, or 3A **Medical Methodology:** Autologous anti-CD19 Chimeric Antigen Receptor (CAR) T-cell Therapy **Optimization Target:** High Complete Response Rate (CRR) and prolonged Duration of Response (DOR)

13. Study Procedures & Methodology

Management Pathway: Leukapheresis → Cell Manufacturing → Lymphodepleting Chemotherapy → Tisagenlecleucel IV Infusion **Education Protocol (HCP):** Specialized site certification required for safe administration of CAR-T cell therapies, including rigorous CRS and neurotoxicity management guidelines. **Education Protocol (Patient):** Education regarding leukapheresis procedures, signs/symptoms of CRS/ICANS requiring immediate medical attention, and prolonged follow-up. **Quality Audit Mechanism:** Centralized Independent Review Committee (IRC) radiological assessments for objective tumor response.

14. Observation & Follow-Up Schedule

Total Duration: $\geq$ 24 months of primary efficacy/safety follow-up; extending to long-term multi-year survival follow-up.
Onsite Visit Cadence: Frequent early visits (Day 28, Month 3) for primary PET/CT restaging and acute safety monitoring. **Remote Monitoring Frequency:** Routine health checks for latent AEs.
Checklist Review Interval: Survival status and disease progression monitored every 3 months after the core follow-up phase.

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Statistician	[Name]	_____	[DD-MMM-YYYY]						